

EREN MENGES

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EDUCATION

B.S. Computer Science, NYU Tandon

Expected May 2028

- Coursework: Design and Analysis of Algorithms, Linear Algebra, Probability/Statistics, Quantum Science
- GPA: 3.3, Minor in Math

SKILLS

ML: PyTorch | numpy | Transformer Models

Systems/Dev: Python (Professional) | C++ (Intermediate) | Linux/Debian | Bash

Research Interests: AI4Science | Graph Neural Networks (GNN) | Spatio-temporal modeling | Physics-informed ML

Languages: **English:** Fluent, **German:** Advanced (TELC B2), **Turkish:** Native

EXPERIENCE

Machine Learning Research Intern, C2SMART Center for Urban Mobility, NYU

Jun 2025 - Aug 2025

- Trained, benchmarked and profiled 7 trajectory prediction models on Argoverse2 to optimize the accuracy-efficiency trade-off for autonomous systems.
- Quantified model performance across diverse metrics (minADE/FDE, miss@2m), establishing a performance baseline of 0.56–0.80m error against 3.4–16ms inference latency.
- Validated the EMP architecture as a high-efficiency surrogate, maintaining MTR-level accuracy while reducing theoretical complexity by ~50% FLOPs and accelerating training throughput by 6× (8h vs 50h).
- Managed cloud GPU training on vast.ai (checkpoints, failure recovery, logs)

PROJECTS

Flood Denoising and Forecasting with STGCNNs

November 2025 - Present

- Developing a Spatio-Temporal Graph Convolutional Network (STGCNN) to model flood dynamics across 229 physical sensor nodes in NYC, capturing both spatial topology and temporal dependencies.
- Engineering a real-time forecasting pipeline to predict water levels 30 minutes ahead, integrating LSTM modules to handle long-term temporal dependencies in noisy sensor data.
- Benchmarking deep learning architectures against classical baselines (XGBoost) to quantify gains in predictive accuracy and noise robustness.

Spectral Clustering with Linear Algebra

December 2025 - February 2026

- Implemented a Spectral Clustering algorithm from scratch to identify non-convex structures often misclassified by classical K-Means.
- Built a custom linear algebra pipeline to compute pairwise distance matrices, build mutual k-NN graphs, and compute Graph Laplacians to project high-dimensional data into separable eigenspaces.
- [Interactive demo.](#)

ElectroTA - Hardware-in the Loop Multimodal Assistant for Circuit Experiments, Co-Creator

March 2026

- Designed a system spanning an ESP32 (live voltage probes + camera over serial), a local Python client, and a Cloud Run server orchestrating Gemini Live API sessions with custom voice activity detection (VAD) and tool calling.
- Architected a CV pipeline combining Gemini's Segmentation API with classical OpenCV to detect components, read resistor color band directions, and identify component polarities from breadboard photos.
- Demonstrated live circuit debugging with real-time voice guidance and wiring error detection.
- [Short Live Demo](#)